

Hridyesh Kumar

Machine Learning / AI Intern Candidate

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SUMMARY

Final-year B.Tech student at IIIT Allahabad targeting a Machine Learning / AI internship at a quantitative firm. Builds and rigorously evaluates models end-to-end — deep learning (PyTorch, CLIP, CNN ensembles) and gradient-boosted models for markets — with a focus on honest evaluation (walk-forward validation, data-leakage audits). Also writes low-latency C++ (a limit-order-book matching engine at ~13M orders/s). Codeforces Expert, LeetCode Knight (top 6%).

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL
ML & Deep Learning: PyTorch, Transformers, scikit-learn, LightGBM, NumPy, Pandas, OpenCV
ML Concepts: CNNs, Transformers, CLIP / Vision-Language Models, Multimodal Fusion, Gradient Boosting, Cross-Sectional Ranking, Walk-Forward Backtesting, Retrieval-Augmented Generation (RAG), Grad-CAM Explainability
Quant & Backend: Feature Engineering, Time-Series Validation, Risk Parity, FastAPI, Docker, Qdrant (Vector DB), Hugging Face, REST APIs
CS Fundamentals: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Computer Networks, DBMS

PROJECTS

- AlphaRank — ML Quant Trading System (NSE)** | *Python, LightGBM, pandas, Flask* GitHub
- Built a systematic equity system that computes ~50 point-in-time features on ~380 NSE stocks and uses a multi-horizon (5/10/20-day) LightGBM model to rank stocks cross-sectionally each week, holding a risk-parity basket of the top 7 with volatility-adaptive stops, sector caps, and a drawdown kill-switch.
 - Validated with a cost-adjusted walk-forward backtest (rolling retrain, horizon-length embargo); **caught and fixed a data-leakage bug** that had inflated the Information Coefficient from a fake 0.48 to an honest ~0.01–0.02, then rebuilt the metric as a per-date Spearman rank IC. Runs on free data in paper-trading mode with a live dashboard.
- Facial Emotion Recognition — RAF-DB** | *PyTorch, OpenCV, Grad-CAM, Gradio* GitHub | Live Demo
- Fine-tuned DDAMFN (a face-specialised attention CNN) on RAF-DB (7 emotions, ~15K images) and built a DDAMFN + ConvNeXt-V2 stacking ensemble reaching **91.6% test accuracy**, on par with published state-of-the-art.
 - Added Grad-CAM explainability, deployed a live Gradio demo on Hugging Face Spaces, and fixed a data-leakage bug for an honest, leak-free train/validation/test split.
- Multimodal Product Tagging — CLIP** | *PyTorch, CLIP, Transformers, Gradio* GitHub | Live Demo
- Built a CLIP-based multimodal model that auto-tags products with 4 catalog attributes from image + title in one forward pass — **92% mean accuracy** (category 98%, gender 99%) across 30K products.
 - Quantified robustness to missing metadata: with titles removed, mean accuracy holds at 82% as the model falls back on CLIP image features; deployed as a live Gradio demo.

Also: Match Engine — price-time-priority limit-order-book matching engine in C++ (~13M orders/s, p99 ~400ns; O(1) cancel, arena allocator, lock-free SPSC feed, 71 tests); **RepoSage** — agentic RAG over codebases with hybrid retrieval (dense + BM25, RRF), reaching MRR 0.81.

EDUCATION

Indian Institute of Information Technology, Allahabad	2023 – present
<i>B.Tech in Electronics and Communication Engineering</i>	<i>CGPA: 7.34</i>
St. Mary's Inter College, Etawah	2019 – 2022
<i>Class XII (CBSE) 94.4% — Class X (CBSE) 97.4%</i>	

CODING PROFILES

Codeforces: Expert (1701) — **LeetCode:** Knight (1863, top 6%) — **CodeChef:** 3-Star (1720)

ACHIEVEMENTS

Top 150 nationally — TenzorX 2026 National AI Hackathon (team).
NTSE Scholar (2020): Top 2000 out of 1.2 million applicants.
PRMO (2018): Top 456 in Uttar Pradesh — **VVM** (2020): Top 18 in Uttar Pradesh.